

Highlights

DaCaF: Bayes-optimal per-metric inference for probabilistic multi-label classification

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- Bayes-optimal per-metric inference for any probabilistic multi-label classifier
- Divide-and-conquer plus fusion: exact optima over two whole metric families
- Seven predict-rules, each verified against brute-force enumeration
- pip-installable and Dockerised, one-command reproduction of paper tables

DaCaF: Bayes-optimal per-metric inference for probabilistic multi-label classification

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ARTICLE INFO

Keywords:

multi-label classification

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evaluation metrics

Code metadata

Nr.	Code metadata description
C1	Current code version: v1.0.0
C2	Permanent link to code/repository: https://github.com/hxtruong6/inference_probabilistic_mlc
C3	Zenodo: TODO software DOI Permanent link to Reproducible Capsule: TODO Code Ocean capsule
C4	Legal Code License: MIT
C5	Code versioning system used: git
C6	Software code languages, tools, services used: Python (≥ 3.10); NumPy, SciPy, scikit-learn, pandas, joblib; optional PyTorch / torchvision / torctxrayvision ([image] extra)
C7	Compilation requirements, operating environments, dependencies: <code>pip install -e . (core) or ".[image]"</code> ; Linux/macOS; Docker image provided
C8	Link to developer documentation/manual: README.md, CONVENTIONS.md, docs/
C9	Support email for questions: hxtruong@jaist.ac.jp

1. Motivation and significance

- Multi-label classification: each instance carries any subset of L labels.
- Key fact: different evaluation metrics are optimised by different predictions; a model great for F_1 can be poor for subset accuracy.
- Right per-instance choice is the Bayes-optimal prediction (BOP): the $\hat{y}$ maximising the *expected* score of the target metric.
- Computing BOP naively is intractable (2^L candidate predictions).

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ABSTRACT

TODO abstract (≤ 250 words).

- Gap: no reusable, exact, multi-metric BOP tool for probabilistic MLC; prior work treats one metric at a time and/or approximates.
- DaCaF closes this: a generic recipe + reference implementation that, given any probabilistic model $P(y | x)$, returns the exact BOP for a chosen metric.
- Significance: optimise the metric you actually care about; study metric-mismatch effects without approximation noise.
- Tie to the published paper (Information Fusion 2026, Nguyen et al., 2026): proves correctness for two metric families; this software is the artifact behind it.

2. Software description

2.1. Software architecture

- Installable Python package `dacaf_mlc` (pip / Docker; console script `dacaf-mlc`).
- Core: `ProbabilisticClassifierChainCustom` (PCC) wraps any scikit-learn base estimator; estimates $P(y | x)$ via a classifier chain.
- Two algorithmic building blocks behind every inference rule:
 - Divide-and-conquer: partition the 2^L predictions into $L+1$ groups by number of relevant labels; best-in-group by sorting labels by a score; global best = best across groups.
 - Fusion: scores need marginal/pairwise probabilities, estimated by fusing the chain's dependent binary classifiers (ancestral sampling).
- Supporting modules: `evaluation_metrics.py`, `metrics_registry.py`, `pipeline.py` (train / k-fold / run_single), `evaluate.py` (CLI), `arff_dataset.py` (MULAN ARFF + 10-fold CV), `datasets.py`, `chest_xray_dataset/` (NIH feature extractor, [image] extra).
- Vendored `skmultiflow` `ClassifierChain` base for a self-contained install.

Table 2

CHD-49, PCC + logistic regression, mean over 5 seeds $\times$ 10-fold CV (% , higher is better). Read each column: the bold diagonal entry (optimise the metric you evaluate) is the maximum of its column. NPV and Recall collapse to the predict-all trivial optimum.

Target ↓ / Eval →	F_1	Hamming	Markedness	NPV	Precision	Recall	Subset
Predict F_1	67.07	69.29	71.91	81.09	62.74	79.22	15.06
Predict Hamming	63.92	70.78	71.27	75.74	66.62	66.93	18.05
Predict Markedness	34.20	63.70	76.96	71.10	33.37	40.45	8.39
Predict NPV	58.35	43.01	71.50	100.00	43.01	99.46	0.00
Predict Precision	40.54	64.83	68.31	63.10	73.52	29.10	3.10
Predict Recall	58.35	43.01	71.50	100.00	43.01	99.46	0.00
Predict Subset	63.96	69.37	70.35	76.47	64.15	69.79	18.92

2.2. Software functionalities

- Seven per-metric Bayes-optimal rules: `predict_fmeasure`, `predict_hamming`, `predict_markedness`, `predict_precision`, `predict_npv`, `predict_recall`, `predict_subset`.
- Each returns the prediction maximising the expected value of its target metric; conventions in `CONVENTIONS.md`.
- Covers two whole metric families at once; flags when a metric's optimum is *trivial* (a metric-choice diagnostic).
- Batched predictor: one `predict_proba` per chain level instead of $N \cdot L \cdot 2^L$; verified numerically equal to brute-force enumeration.
- Reproducible evaluation: CLI + scripts (`reproduce_tabular.sh`, Slurm) over MULAN tabular datasets and an NIH ChestX-ray14 image dataset.
- Correctness harness: every rule checked against brute-force expected-metric enumeration on small cases; CI on Python 3.10/3.12.

3. Illustrative examples

- Minimal library usage (fit once, query any metric's BOP):

```
from sklearn.linear_model import LogisticRegression
from dacaf_mlc.probability_classifier_chains \
    import ProbabilisticClassifierChainCustom
from dacaf_mlc.evaluation_metrics import EvaluationMetrics as EM

pcc = ProbabilisticClassifierChainCustom(
    LogisticRegression(max_iter=10_000))
pcc.fit(X_train, Y_train) # Y: (n, L) binary
y_f1 = pcc.predict_fmeasure(X_test, beta=1) # BOP for F1
y_ham = pcc.predict_hamming(X_test) # BOP for Hamming
print(EM.f_beta(Y_test, y_f1), EM.hamming(Y_test, y_ham))
```

- Headline experiment (mismatch hurts): Table 2, rows = metric optimised for, columns = metric evaluated; bold diagonal = column maximum.
- One command reproduces it: `make reproduce` (CHD-49, fastest dataset).
- The chest-xray path scales the same API to real DenseNet/ResNet features.

4. Impact

- Enables an exact research question: how much does optimising the wrong metric cost, measured without approximation blurring the picture?
- Model-agnostic: works with *any* probabilistic MLC model; a drop-in BOP layer on top of existing $P(y | x)$ estimators.
- Breadth: one tool covers two whole metric families; the trivial-optimum diagnostic guides metric selection.
- Reuse beyond the paper: clinical/image MLC (NIH ChestX-ray14 demo), benchmarking, and teaching Bayes-optimal decision-making in MLC.
- Trust: closed-form rules verified against brute-force enumeration; CI keeps it reproducible.
- Low barrier: pip-installable, Dockerised, scripted reproduction, MULAN datasets bundled; a newcomer reproduces a paper table in minutes.
- Extensibility: registry pattern to add metrics/datasets; vendored base keeps installs self-contained.
- Who benefits: MLC researchers, applied ML practitioners choosing a deployment metric, and educators.

5. Conclusions

- DaCaF turns a proved theoretical recipe into a reusable, verified, exact per-metric inference tool for probabilistic MLC.
- Future: more base models, more metric families, optional approximate fast paths.

Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

TODO statement (or remove if not applicable).

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